High-Resolution Feature Generator for Small-Ship Detection in Optical Remote Sensing Images

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***Abstract*— Ship detection in optical remote sensing (RS) images remains a persistent challenge in current research. While prevailing methods achieve satisfactory outcomes in detecting large ship objects within RS images, the identification of small-ship objects poses greater difficulty due to their limited pixel information. To address this challenge, the utilization of generative adversarial network-based (GAN-based) super-resolution (SR) techniques proves effective. Therefore, in this article, we present a high-resolution feature generator (HRFG) specifically tailored for small-ship detection. Different from previous GAN-based methods that rely on image-level SR or feature sharing between SR and detection, we design a new architecture that uses an additional network branch, that is, high-resolution feature extractor (HRFE), to extract real high-resolution (HR) feature as a feature-level supervisory signal. The intuition is that real HR features may guide the generator network to extract HR features from low-resolution (LR) images directly. Consequently, the feature for detection is extracted and enhanced at the same time so that a large amount of calculation brought by image-level SR is avoided. Additionally, we introduce a background degradation strategy within the HRFE to improve the performance of small object recognition. Extensive experiments on a self-assembled ship dataset and two other public datasets show the superiority of the proposed method in small-ship detection tasks.**

***Index Terms*— Generative adversarial network (GAN), optical image, remote sensing (RS), ship detection.**

I. Introduction

THE utilization of ship detection technology in optical remote sensing (RS) images presents various opportunities across domains including environmental regulation, surveillance, military, and traffic management [1]. In recent years, the rapid advancement of deep learning has significantly impacted RS image processing [2], [3], [4], [5], which also benefits ship detection techniques a lot. For example, leveraging the success of convolutional neural network (CNN)-based detectors, several studies have integrated CNN techniques into RS ship detection tasks and got amazing results [6], [7], [8], [9], [10], [11], [12], [13], [14], [15], [16].

However, the existing ship detection methods perform unsatisfactorily for low-resolution (LR) images, especially when the ship objects are small. Even if on high-resolution (HR) images, the detection performance for small ships is lower than that for large ships [17]. This is because small ships only cover a small area in RS images and lack discriminative features. Previous research studies have acknowledged the complexity of detecting small objects and have proposed numerous related strategies, such as multiscale feature fusion [18], sample-oriented [19], spatial attention [20], and so on. Because a large amount of information about small targets will be lost after the feature is pooled multiple times, the multiscale feature fusion strategy enhances the multiscale features and fuses the contextual semantic information between them. Sample-oriented strategy believes that the performance of deep learning depends on a large amount of training data in a sense, thus the performance of small object detection can also be improved by increasing the category and number of small target samples in the training set. The spatial attention strategy utilizes an attention mechanism to increase target hints and weaken nontarget information. Despite all their effectiveness to some extent, these strategies fail to fundamentally address the issue of lacking features. Presently, small-object detection remains a challenging problem within computer vision. A straightforward way to alleviate the problem mentioned above is by enriching the features of small objects by mimicking those of larger ones. To this end, researchers have used super-resolution (SR) techniques to generate artificial images or features for subsequent object detection and some studies achieved noticeable improvements when using the SR images for detection [21], [22].

Parallel to advancements in object detection, deep-learning-based image SR has seen considerable success [23], [24], [25], [26], [27], notably through the utilization of generative adversarial networks (GANs) renowned for their image-generation capabilities [28], [29], [30], [31], [32]. Consequently, numerous GAN-based object detection methods have emerged, all of which aim to super-resolve objects in images to ease the difficulty of detection. These GAN-based detection models typically comprise three subnetworks: an SR generator, a discriminator for generator training, and an object detection module. However, some methods [21], [22], [33], [34] simply concatenate an SR module with detection networks, maintaining relative independence. The SR module incurs significantly greater computational demands

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than object detection due to increased feature dimensions and computational load, resulting in prolonged processing times, especially on extensive RS images. As a solution, [35], [36], [37], and [38] utilize two-stage detection networks to cut regions of interest (RoIs) of objects and generate SR ROIs instead of the whole image. This strategy significantly improves the detection speed but region proposal networks (RPNs), which cut the patches of RoIs, limiting the performance of the network.

To sum up, in detection tasks, SR images are not necessary and the SR feature is what we essentially want to obtain. Consequently, prioritizing feature-level SR over image-level SR appears more logical, and the most critical issue is to obtain ground-truth HR features as supervisory signals. Furthermore, optical RS images are usually more degraded and contain complex backgrounds. Even in HR optical RS images, intricate land environments, large areas of wave ripple, or docks closed to ships seriously interfere with ship detection.

To solve the above problems, in this article, we propose an HR feature generator (HRFG)-based method to better detect the small ships in optical RS images. The proposed architecture is depicted in Fig. 1, which contains a high-resolution feature extractor (HRFE) to extract ground-truth HR features as a supervisory signal during training, an HRFG to extract HR features from LR images directly, a binary classification discriminator to train the HRFG, and a detector to detect ships. Different from previous methods that use indirect supervision [39] or share feature extraction network [36], we design an additional network branch, that is, HRFE to obtain HR feature for direct supervision.

Considering that the huge background may dominate the training phase of generating HR features and raise the difficulty of ship identification, we introduce a background degrade block (BDB) to blur the background and keep ship objects clear. Our intuition is that by blurring the background, features extracted by HRFE will contain more information about ships, thus guiding the HRFG to focus more on ships instead of the background. The strategy of degrading background is inspired by Chen et al. [7] and we simplify its method to ensure clarity of multiscale ships and reduce training costs.

We use YOLOv8s6 [40] as the baseline and our network primarily comprises its backbone, neck, and head structures. During the inference phase, our network is exactly the same as the baseline except for an extra upsampling layer at the input. We wish our proposed method could be implemented on most CNN-based one-stage detectors by this design.

To train our model, we collected data from two HR optical RS image datasets, that is, HRRSD [41] and HRSC2016 [42] to form our HR dataset Ship-HR. Then, we downsampled its HR images to obtain the LR dataset Ship-LR. During the training phase, HRFE extracts HR features from Ship-HR and guides HRFG to generate HR features from Ship-LR. In addition, we note that in practical applications, we usually have images with small ships but lack corresponding HR images. Hence, we also transfer our model to two public optical RS small-ship detection datasets, that is, MASATI [43] and LEVIR-Ship [7], to analyze the generalization ability of our model. The results show the effectiveness and efficiency of our method.

The contributions of our work can be summarized as follows.

1) We propose an HRFG-based ship detection method on optical RS images. Unlike the previous GAN-based detection methods, we utilize an extra branch network named HR feature extractor (HRFE) to obtain direct supervisory signals during training.
2) We designed a new and simple background degradation method to guide the generator to generate features that focus more on the ship objects.
3) We extend the feature-level strategy to a one-stage detector. Extensive experiments on the self-assembled dataset and two public datasets validate the effectiveness of the proposed method.

The rest of this article is organized as follows. Related work is introduced in Section II. In Section III, we give a detailed description of our method. Experimental results are reported in Section IV and the conclusion is drawn in Section V.

II. Related Works

A. Object Detection

Traditional object detection methods make use of hand-crafted features to locate the objects of interest from backgrounds [44], [45], [46]. Since the manually designed features are vulnerable to complex situations, these methods suffer from serious misdetection problems.

In recent years, object detection algorithms have basically been implemented through deep learning, and many effective CNN-based detectors have been proposed. They are very robust in complex environments and can be divided into one- and two-stage methods. Two-stage methods are represented by the region based convolutional neural network (RCNN) series [47], [48], [49]. They first obtain candidate bounding boxes and then classify and regress the results from the candidate. Compared with two-stage methods, one-stage methods represented by YOLO [50], [51], [52], SSD [53], and EfficientDet [54], [55] are designed to directly give the detection results. However, the methods mentioned above rely on the prior setting of anchor boxes which constrains their generalization ability. Hence, a series of anchor-free methods are proposed, for example, CenterNet [56], FCOS [57], and YOLOv8 [40]. They enjoy larger and more flexible solution space and eliminate the computational burden associated with using anchors, thereby pushing both detections further toward real-time and high precision.

B. Ship Detection in Optical RS Images

Ship detection has been an active research topic in the field of RS for decades. Inspired by CNN-based detectors, many improved methods for ship detection tasks based on general object detectors have been proposed. Li et al. [58] proposed HSF-Net based on the Faster-RCNN [48]. It uses a hierarchical selective filtering layer to map features in different scales to the same scale space. Tian et al. [59] designed an image enhancement module driven by object characteristics and implemented it on YOLO and Mask-RCNN [49]. Chen et al. [60] designed a novel and lightweight dilated attention module and embedded it into the basic YOLOv3. Zhang et al. [61] proposed an efficient channel attention model with the self-attention mechanism

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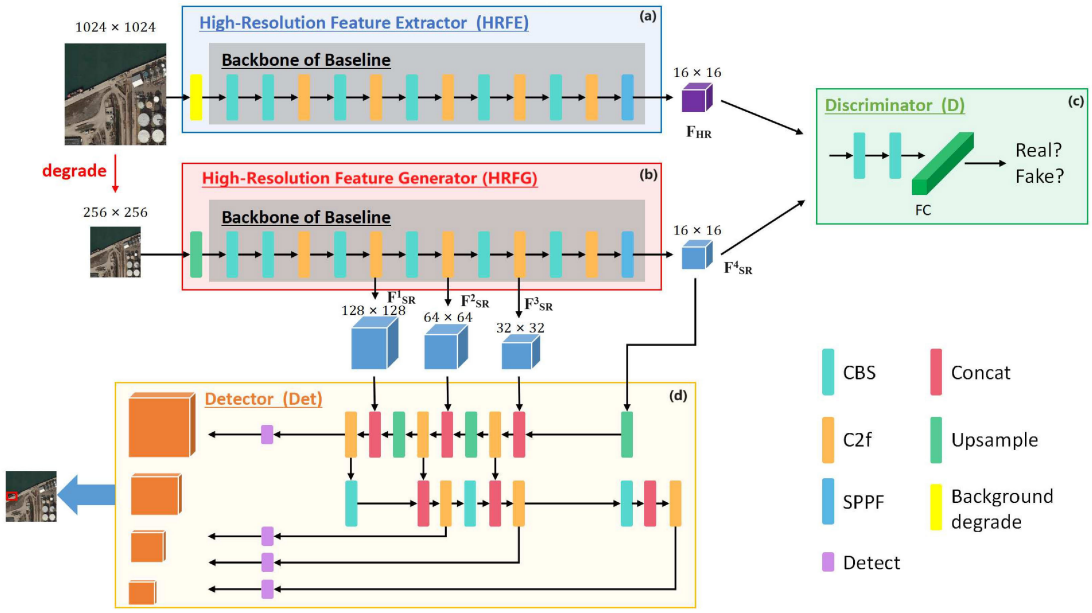


Fig. 1. Overall network architecture. (a) HRFE: a series of BDB and pretrained backbone. It is designed to extract HR features as supervisory signals. The backbone is the same as YOLOv8s6; so, for details of CBS, CSP bottleneck (C2f), and SPPF, refer to [40]. (b) HRFG: a series of quadruple upsample layers and backbone. It is designed to extract HR features from LR images directly. (c) Discriminator (D): a simple binary classification network that consists of two convolution blocks and fully connected layers. (d) Detector (Det): the neck and head of the baseline.

to enable information interaction between channels without adding additional computational overhead. Chen et al. [7] introduced an additional degraded reconstruction enhancer, which intended to regress an object-aware blurred version of the input image during the training phase. Additionally, they integrated a cross-stage multihead attention module into the detector, leveraging the self-attention mechanism to enhance feature discrimination further. Wen et al. [62] discussed the issues of SSD on multiscale objects and proposed a multiscale single-shot detector (MS-SSD) to both improve the detection accuracy of small-ship targets and the model's robustness to scale variance.

In fact, many current ship detection methods are experimented on high-resolution datasets, where the details of the target are clear to achieve high performance. when facing the tiny ship detection problem, they may suffer performance degradation. In this article, we aim to address the tiny ship detection problem.

C. GAN-Based Enhancement for Detection

Utilizing the GAN-based SR technique to enrich objects' features is a feasible way to tackle tiny ship detection problems, and GAN-based detection methods could be divided into image-level SR-based methods and feature-level SR-based methods. At present, the former methods have more flexibility to replace their SR module but inevitably generate artificial textures and artifacts, which adversely affect the accuracy of detection. The latter has less computational burden but has difficulty obtaining ground-truth HR features as supervisory signals. In 2017, Li et al. [39] realized that the generation ability of GAN could be applied to small object detection tasks and first introduced GAN into detection. However, limited by the indirect supervision strategy, the performance of the network has not improved significantly. Then, Bai et al. [35] propose an end-to-end multitask GAN (MTGAN). In the MTGAN, the generator is an SR network, which can upsample small blurred images into fine-scale ones and recover detailed information for more accurate detection. The discriminator is a multitask network, which describes each SR image patch with a real/fake score, object category scores, and bounding box regression offsets. Noh et al. [36] proposed a novel feature-level SR approach that is integrable with any proposal-based detectors with feature pooling. These GAN-based methods focus on daily natural images which contain relatively simple backgrounds. Rabbi et al. [33] applied a new edge-enhanced SR GAN (EESRGAN) to improve the quality of RS images and used different detector networks in an end-to-end manner where detector loss was backpropagated into the EESRGAN to improve the detection performance.

In general, there are not many studies on small target detection methods based on GANs. The key problem that such methods need to solve is how to avoid the huge time cost brought by SR. In this article, we proposed a novel HRFG-based architecture, which belongs to the feature-level SR category and could be implemented on a one-stage detector to address small-ship detection.

III. Proposed Method

The overall network architecture of our model is shown in Fig. 1. It consists of an HRFE, an HRFG, a discriminator (D), and a detector (Det). We follow YOLOv8s6 (a version of YOLOv8) [40] as the baseline. The baseline has three parts, that is, a backbone, a neck, and a head. The HRFE incorporates a BDB in tandem with the backbone, specifically designed for the extraction of real HR features serving as supervisory signals. The backbone of the HRFE is trained on the HR ship detection dataset and then frozen in training. The HRFG is designed to extract HR features from LR images directly, which is composed of a quadruple upsample layer and the backbone in series. D is a simple binary classification

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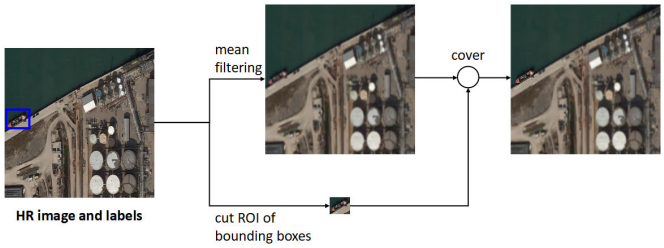


Fig. 2. Specific operation of the BDB: use labels of the HR image to cut the ROIs of bounding boxes and use mean filtering to blur the whole image. Finally, cover the blurry objects with the clear ROIs.

network that consists of two convolution blocks and fully connected layers. It encourages HRFG to generate realistic features similar to HR features. Det is the neck and head of the baseline with no changes. For clarity in the subsequent context, we will refer to the HR features generated by HRFG as SR features. Further details of our model will be elaborated subsequently.

A. HR Feature Extractor

The HRFE directly extracts features of the whole HR image and uses the extracted HR features as a supervisory signal to guide the HRFG to generate SR features for detection. However, because small ships only account for small areas in an image, most of the features extracted by HRFE belong to the complex background and only a small portion belong to ship objects. Features of a vast background would dominate the training phase of the HRFG, making it difficult to generate SR features and recognize ship objects. Hence, inspired by Chen et al. [7], we add a BDB to let the HRFE obtain HR features with blurry backgrounds and clear objects, which would guide the HRFG to pay more attention to ships.

The specific operation of the BDB is shown in Fig. 2. Different from [7], which utilizes the features extracted by the backbone to reconstruct an object-aware blurred version of the input image, our proposed BDB takes labels of HR image as supervision to omit the background and keep more information about the ships. It first cuts ROIs of bounding boxes from the HR image to get clear ship targets. Second, it uses a 17×17 mean filter to blur the HR image, creating a degraded background. To keep the objects clear, the blurry objects are covered with the clear ROIs finally.

B. Generator Network

The generator network HRFG is designed to extract discriminative features from LR images directly, which follows the backbone of YOLOv8s6 to couple the detector better. To resolve scale discrepancies between real HR and generated SR features, a quadruple upsample layer is introduced at the backbone's input, employing bicubic interpolation within the HRFG.

Let F be the features extracted and I be the input images. The HRFG has four output layers which, respectively, output four groups of feature maps $\{F_{\mathrm{SR}}^{i}, i = 1, 2, 3, 4\}$ from LR image I_{LR}. Due to the superior feature expression capacity of deep networks over shallow ones, shallow-generated SR feature maps cannot align as precisely with real HR feature maps as deeper representations. Consequently, only the deepest feature maps F_{SR}^{4} are regressed with corresponding real HR feature maps F_{HR} extracted by HRFE. When F_{SR}^{4} is utilized to calculate the loss, all layers of the G will be trained.

Notably, HRFG extracts feature maps for detection and super-resolves them at the same time, avoiding using an extra generator network or backbone of the detection model. Therefore, this approach improves efficiency by combining detection-related feature extraction and SR processes within a single network structure.

C. Discriminator Network

Our discriminator (D) is utilized to employ adversarial training to ensure that the features generated by HRFG closely resemble those extracted by HRFE. Considering that D uses extracted feature maps from a deep backbone, employing elaborate networks like VGG [63] or GoogLeNet [64] becomes unnecessary. Our discriminator network D adopts a simplified VGG architecture, comprising two convolutional blocks (CBS) and fully connected layers with a sole hidden layer.

In our approach, we utilize the relativistic discriminator proposed in [65] to train HRFG. Consequently, the sigmoid activation function within the output layer is omitted. The discriminator attempts to predict the probability that measures whether a real feature F_{HR} is relatively more realistic than a generated feature F_{SR}, as follows:

$$D(F_{\mathrm{HR}}, F_{\mathrm{SR}}) = \sigma(C(F_{\mathrm{HR}}) - \mathbb{E}[C(F_{\mathrm{SR}})]) \to 1 \tag{1a}$$

$$D(F_{\mathrm{SR}}, F_{\mathrm{HR}}) = \sigma(C(F_{\mathrm{SR}}) - \mathbb{E}[C(F_{\mathrm{SR}})]) \to 0 \tag{1b}$$

where $D(\cdot, \cdot)$represents the final probability, σ denotes the sigmoid function, $C(\cdot)$ represents output of the discriminator, and $\mathbb{E}(\cdot)$ is the mathematical expectation.

D. Detector Network

The anchor-free detector network Det follows the neck and head of YOLOv8s6. It predicts the distance distribution from the cell center to the four boundaries of the bounding boxes, as well as the distance inferred from the distribution's expectation. Compared with the baseline, our model remains unchanged, featuring only an additional upsampling layer at the input during detection tasks. Consequently, we wish our approach to readily extend to other one-stage detection models. For details of Det, refer to [40].

E. Degradation of the HR Image

In the existing GAN-based detection methods [33], [35], [36], [39], pairwise HR-LR images are obtained by downsampling HR images in a fixed way. However, the degradation mode of RS images in the real world is complex and random. Hence, inspired by Cui et al. [66], our degenerate method introduces random Gaussian blur and Gaussian noise at the same probability 30% and uses bicubic downsample with factor 4 to get LR images. For the blur operation, we set the Gaussian kernel size to be uniformly sampled from {3,5,7,9,11,13,15,17}. For the noise operation, we adopt a zero-mean additive white Gaussian noise (AWGN) model. The variance σ is randomly chosen from a uniform distribution $U(0, 25/255)$.

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F. Loss Function

The loss function is composed of three parts, including the detector loss, the generator loss, and the discriminator loss.

1) Detector Loss: The detector loss function follows the loss function in YOLOv8s6, which contains regression loss, classification loss, and distribution focal loss (DFL). It conducts typical loss in object detection task—complete intersection over union (CIoU) loss [67] for regression error and binary cross-entropy loss for classification. They are defined as

$$\text{loss}_{\text{reg}} = 1 - \text{CIoU} \tag{2}$$

$$\text{CIoU} = \text{IoU} - \frac{\rho^2(b, b_{\text{gt}})}{c^2} - \frac{v^2}{1 - \text{IoU} + v} \tag{3}$$

$$v = \frac{4}{\pi^2}\left[\left(\arctan\frac{w_{\text{gt}}}{h_{\text{gt}}} - \arctan\frac{w}{h}\right)^2\right] \tag{4}$$

where b and b_{gt} denote the central points of the predicted box and the target box, $\rho(\cdot)$ denotes the Euclidean distance, c is the diagonal length of the smallest enclosing box covering the two boxes, v measures the consistency of the aspect ratio, and w_{gt}, h_{gt}, w, and h, respectively, represent the width and height of the two boxes

$$\text{loss}_{\text{cls}} = -\frac{\sum\left(y_n \cdot \ln y_n^* + (1 - y_n) \cdot \ln\left(1 - y_n^*\right)\right)}{n} \tag{5}$$

where n represents the number of all cells of prediction grid and y_n^* denotes the probability of the nth predicted bounding box. y_n is the label corresponding to the predicted bounding box, which is formulated by CIoU. Since a specific expected value of distance corresponds to an infinite number of distance distributions, DFL is used to center the distribution around the true distance. For specific calculations of DFL, refer to [40] and [68]. Finally, the detector loss is designed as

$$\text{loss}_{\text{det}} = 7.5 \times \text{loss}_{\text{reg}} + 0.5 \times \text{loss}_{\text{cls}} + 1.5 \times \text{loss}_{\text{dfl}} \tag{6}$$

2) Generator Loss: The generator loss function consists of the detector loss, pixel loss, and GAN loss. Our primary objective is to attain high detection performance rather than generating realistic SR feature maps. Consequently, leveraging Det to constrain HRFG effectively becomes crucial. Moreover, since the generator loss contains the detector loss, we could optimize the Det while training the HRFG. In object detection tasks, pixel loss represents a common loss function, manifesting as the L_2 loss between SR feature maps and HR feature maps

$$\text{loss}_{\text{pix}} = \|F_{\text{HR}} - F_{\text{SR}}^4\|_2^2. \tag{7}$$

GAN loss is the generator part of relative discriminator loss

$$\begin{aligned}\text{loss}_{\text{gan}}^G = &-\mathbb{E}_{F_{\text{HR}}}\left[\ln\left(1 - D\left(F_{\text{HR}}, F_{\text{SR}}^4\right)\right)\right] \\ &- \mathbb{E}_{F_{\text{SR}}^4}\left[\ln\left(D\left(F_{\text{SR}}^4, F_{\text{HR}}\right)\right)\right].\end{aligned} \tag{8}$$

Finally, the generator loss is defined as

$$\text{loss}_G = 2 \times \text{loss}_{\text{det}} + \text{loss}_{\text{pix}} + 0.5 \times \text{loss}_{\text{gan}}^G. \tag{9}$$

3) Discriminator Loss: The discriminator loss function is the discriminator part of the relative discriminator loss

$$\begin{aligned}\text{loss}_D = &-\mathbb{E}_{F_{\text{HR}}}\left[\ln\left(D\left(F_{\text{HR}}, F_{\text{SR}}^4\right)\right)\right] \\ &- \mathbb{E}_{F_{\text{SR}}^4}\left[\ln\left(1 - D\left(F_{\text{SR}}^4, F_{\text{HR}}\right)\right)\right].\end{aligned} \tag{10}$$

Algorithm 1 Our Algorithm Pipeline

Define: I_{HR}, HR images in a minibatch
Define: B, all the ground-truth boxes set in I_{HR}
Define: I_{LR}, LR images obtained by degenerating I_{HR}
Define: $HRFE(\cdot,\cdot)$, output of high-resolution feature extractor
Define: $Degrade(\cdot)$, a function to degrade image
Define: $HRFG(\cdot)$, output of generator
Define: $Det(\cdot)$, output of detector
Define: $C(\cdot)$, output of discriminator
(1).Training Pipeline:
Input: I_{HR}, B
Output: T, result of object detection on I_{LR}
Output: P_{rf}, probability that HR feature maps are more real than SR feature maps
Output: P_{fr}, probability that SR feature maps are more real than HR feature maps

Freeze HRFE
$F_{HR} = HRFE(I_{HR}, B)$
$I_{LR} = Degrade(I_{HR})$
Freeze D, unfreeze HRFG and Det
$\{F_{SR}^i, i = 1, 2, 3, 4\} = HRFG(I_{LR})$
$T = Det(\{F_{SR}^i\})$
$C(F_{HR}), C(F_{SR}^4) \longrightarrow P_{rf}, P_{fr}$
Backforwards
Freeze HRFG and Det, unfreeze D
$\{F_{SR}^i, i = 1, 2, 3, 4\} = HRFG(I_{LR})$
$C(F_{HR}), C(F_{SR}^4) \longrightarrow P_{rf}, P_{fr}$
Backforwards

(2).Testing Pipeline:
Input: I_{LR}
Output: T

Freeze HRFG and Det
$\{F_{\text{SR}}^i, i = 1, 2, 3, 4\} = HRFG(I_{\text{LR}})$
$T = Det(\{F_{\text{SR}}^i\})$

G. Implementation Details

The size of the input HR images, LR images, and HR feature maps are 1024×1024, 256×256, and 16×16, respectively. To get a powerful HRFE to train the HRFG, we first train the whole baseline on the HR dataset for 300 epochs and save its parameters. When training the HRFG, we use the saved backbone's parameters to initialize the HRFE and take the output of the HRFE to supervise the HRFG. We adopt the stochastic gradient descent (SGD) optimizer with a momentum of 0.99 and a weight decay of 0.0005. The batch size is 8. The initial learning rate of the HRFG, Det, and D is set to 0.001, 0.01, and 0.0005, respectively. And they decay in a cosine annealing strategy until 400 epochs.

Our algorithm pipeline is illustrated in Algorithm 1. After HR features F_{HR} are extracted from I_{HR} by the HRFE, then the degraded version I_{LR} is input into the HRFG to extract SR features F_{SR}. Subsequently, F_{SR} is fed into DT to compute the detection results. Finally, both F_{HR} and F_{SR} are input into D separately to compute the discrimination results. Notably, the HRFG and Det are jointly optimized in the training phase.

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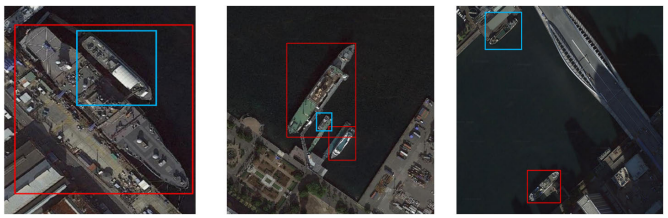


Fig. 3. Supplement of annotations. Red boxes are original labels and blue boxes are our supplement.

In the testing phase, the HRFE and D are removed, and only the HRFG and Det is used. The forward propagation process of the model during testing is relatively straightforward: input the test image into the HRFG, and the output of the HRFG is directly fed into DT, which outputs the detection results.

IV. Experiments

In this section, we conduct experimental comparative analyses of the proposed method. The definition of a small object in this article is an object whose bounding box area is less than 1000 pixels. Our experiment is based on a self-assembled ship dataset. To test the generalization ability of the model, the trained model is transferred to the other two public small-ship detection datasets. All experiments are implemented on an NVIDIA RTX 2080Ti GPU.

A. Evaluation Metrics

We evaluate the effectiveness and efficiency of the proposed method by detection accuracy and model complexity. The metrics are average precision (AP), floating-point operations (FLOPs), model parameters, and frames per second (FPS).

To evaluate detection accuracy, we use AP_{50}, $AP_{50:95}$, and AP_{50}^{small} as the evaluation metric to measure the detection accuracy of the model. In calculating the AP value, we conduct the method that 101 points are sampled at equal intervals in the precision–recall curve according to classification confidence, and then, the precision values corresponding to these points are accumulated and averaged to obtain the final AP value. We use FLOPs, parameters, and FPS to reflect the detection speed.

B. Dataset

As mentioned above, training the model needs an HR ship detection dataset. Hence, we choose two optical RS detection datasets—HRRSD [41] and HRSC2016 [42] (details are shown in Table I). We select images containing ship targets and corresponding annotations from the two datasets to form a new dataset named Ship-HR. What is more, since there are missing annotations in the original dataset, especially the smaller ship objects, we supplement them as shown in Fig. 3. Ship-HR is randomly divided into the training set, the validation set, and the test set, according to the ratio of 7:1:2. Then, we degrade HR images in Ship-HR and obtain the Ship-LR dataset. It should be noted that the data division in Ship-HR and Ship-LR is completely consistent. Details of Ship-HR and Ship-LR are shown in Table II. We add 712 additional annotations and utilize Ship-HR with a supplement to create Ship-LR. In the following experiment, only the datasets with supplements are used. In the Ship-LR, 60.41% of objects are small objects, which brings many challenges to the detection network.

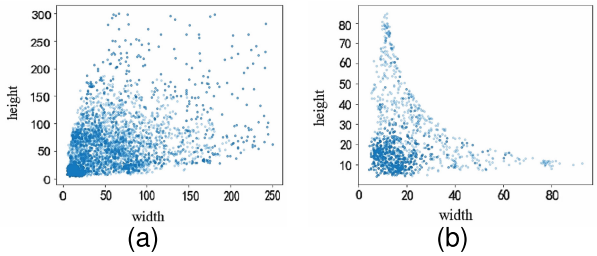


Fig. 4. Ship object size statistics in Ship-LR. (a) All objects. (b) Small objects.

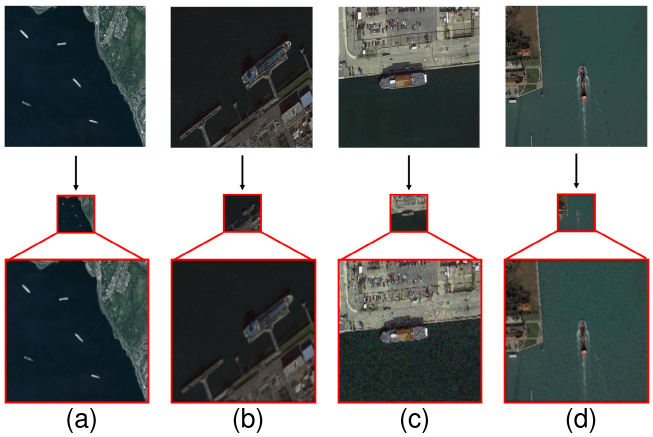


Fig. 5. Partial display of the data contained in Ship-LR under different degradation. (a) Only downsample. (b) Downsample with blur. (c) Downsample with noise. (d) Downsample with blur and noise.

As shown in Fig. 4, we plot the distribution of the width/height of ships in the Ship-LR. To facilitate the observation of the distribution of most target sizes, a few targets with sizes bigger than 300×300 are removed in Fig. 4(a). We can observe that there are multiscale ship targets in the Ship-LR, and the pixel size of small ships is almost below 30×30 and centralizes at around 15×15. A partial display of the data contained in Ship-LR under different degradation is shown in Fig. 5. As described in Section II, the probability of adding noise and Gaussian blur during degradation is 30% and 30%, respectively. Hence, LR images can be divided into four types as Fig. 5.

Moreover, we select two additional public datasets —MASATI [43] and LEVIR-Ship [7], to verify the generalization ability of our model. Most of the objects in MASATI and all of the objects in LEVIR-Ship are small objects. Details of MASATI and LEVIR-Ship are shown in Table III. We use all data of the two datasets except those in the "details" subclass of MASATI and randomly divide the two datasets with the ratio 7:1:2.

C. Comparison

We conducted a comparative analysis of our method against several networks for ship detection using the Ship-LR dataset, including EESRGAN-Det [33], AERIS [66], DRENet [7], EfficientNetV2s [55], and YOLOv8s6 [40]. EESRGAN-Det is tailored for RS detection, comprising a complete SR and detection network. AERIS and DRENet both focus on image reconstruction, with AERIS designed for natural

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TABLE I

TWO PUBLIC HR OPTICAL RS OBJECTS DETECTION DATASETS. HRRSD CONTAINS MULTIPLE CATEGORIES, ONLY INFORMATION ABOUT THE SHIP CATEGORY IS LISTED HERE

Dataset	Images	Images with ship	Objects	Source	Resolution
HRRSD	21761	2773	3886	Google Earth and Baidu Earth	0.15-1.2m
HRSC2016	1070	1070	2976	Google Earth	0.4-2m

TABLE II

SELF-ASSEMBLED SHIP DATASET. OBJECTS WITH A BOUNDING BOX AREA OF LESS THAN 1000 PIXELS ARE CONSIDERED SMALL OBJECTS. "*" REPRESENTS "WITHOUT SUPPLEMENT." IN THE FOLLOWING EXPERIMENT, ONLY THE DATASETS WITH SUPPLEMENT ARE USED

Dataset	Images	Images with ship	Objects	Small objects	Proportion of small objects	Resolution
Ship-HR*	3843	3843	4994	16	0.32%	0.15-2m
Ship-HR	3843	3843	5706	38	0.67%	0.15-2m
Ship-LR	3843	3843	5706	3447	60.41%	0.6-8m

TABLE III

TWO PUBLIC OPTICAL RS SMALL-SHIP DETECTION DATASETS

Dataset	Images Number	Images with ship	Objects	Source	Resolution	Image size
MASATI	5600	2163	3719	Bing Map	unknow	512×512
LEVIR-Ship	3896	2217	3219	GaoFen-1 and GaoFen-6	16m	512×512

TABLE IV

ACCURACY COMPARISON WITH SOME DETECTION METHODS ON SHIP-LR. THE HIGHEST AP VALUE IS MARKED IN BOLD

Method	AP_{50}	$AP_{50:95}$	AP_{50}^{small}
EESRGAN-Det	91.75%	75.05%	79.30%
AERIS	92.17%	76.81%	79.59%
DRENet	86.44%	68.21%	75.08%
EfficientNetv2s	93.20%	74.79%	77.42%
YOLOv8s6(Baseline)	90.84%	74.27%	77.18%
Ours	**95.17%**	**81.99%**	**81.45%**

TABLE V

ACCURACY COMPARISON WITH SOME DETECTION METHODS ON PUBLIC DATASETS. THE HIGHEST AP VALUE IS MARKED IN BOLD

Dataset	Method	AP_{50}	$AP_{50:95}$
MASATI	EESRGAN-Det	74.07%	30.41%
	AERIS	79.18%	38.86%
	DRENet	77.60%	34.29%
	EfficientNetV2s	82.70%	40.32%
	YOLOv8s6(Baseline)	78.39%	39.38%
	Ours	**83.03%**	**40.95%**
LEVIR-Ship	EESRGAN-Det	64.13%	23.73%
	AERIS	72.03%	29.64%
	DRENet	81.35%	37.84%
	EfficientNetV2s	74.40%	31.52%
	YOLOv8s6(Baseline)	72.16%	30.04%
	Ours	**83.73%**	**40.45%**

images and DRENet specifically developed for RS images. EfficientNetV2s and YOLOv8s6 are general-purpose detection networks. Our method, along with EESRGAN-Det, needs paired HR-LR data for self-supervised training and thus both are trained using the Ship-HR dataset. Conversely, the remaining methods are trained using Ship-LR data. All these methods are evaluated using the Ship-LR dataset.

1) Accuracy: Table IV presents a comparative analysis of network detection accuracy using the Ship-LR dataset. Notably, our proposed method attains the highest detection accuracy among other methods for both multiscale and small-ship objects. Specifically, our method establishes a substantial 1.97 point gap in AP_{50} compared to the nearest competitor, EfficientNetV2s. Moreover, it exhibits a noteworthy 4.33% enhancement in AP_{50} over the baseline YOLOv8s6. These results serve as evidence highlighting the significant improvement in our proposed method's performance in detecting small ships.

Fig. 6 visualizes some detection results between ours and other methods. True positive detections are depicted by blue boxes, false negative detections by red boxes, and false positive samples by green boxes. The first row displays results on a heavily degraded image where ships are obscured by significant noise, leading to confusion with the background. In this scenario, our method effectively identifies all ships while minimizing false alarms. Particularly, our method exclusively detects the ship in the lower right corner, a task unaccomplished by other methods. Moving to the second row, the results are showcased on an image featuring a complex coastal background and small ships. Our method demonstrates adeptness in handling intricate land environments. The third row displays results concerning multiscale ships, where smaller ships are situated in proximity to larger ones. Our method adeptly detects these smaller ships attached to larger entities.

Next, we transfer the aforementioned models to the MASATI and the LEVIR-Ship datasets. Since both datasets lack corresponding HR images for small objects, we employ a specific approach for our method and EESRGAN-Det: connecting the generator and detector in series for end-to-end fine-tuning, while maintaining the generator frozen until the penultimate epoch. Each model undergoes fine-tuning for 200 epochs. Table V showcases the comparative detection accuracy across both datasets. Our proposed method outperforms other methods, achieving the highest AP on both MASATI and LEVIR-Ship datasets. Notably, the AP of

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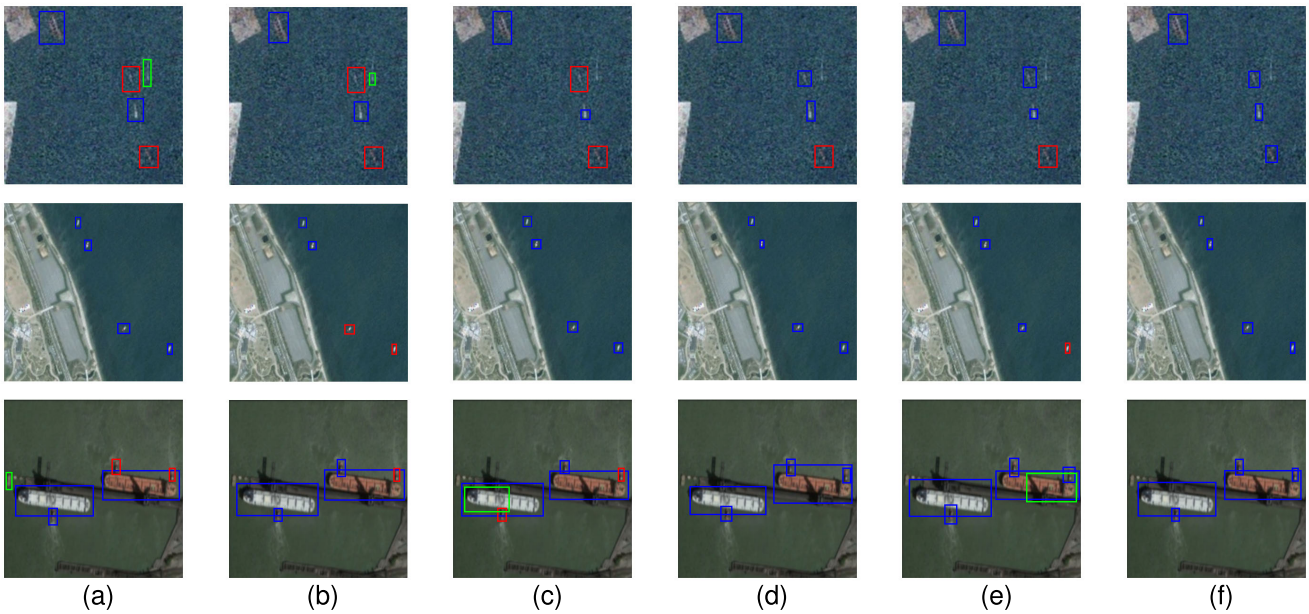


Fig. 6. Comparisons of the detection results on the Ship-LR by different methods. (a) EESRGAN-Det. (b) AERIS. (c) DRENet. (d) EfficientNetv2s. (e) YOLOv8s6 (baseline). (f) Ours. The blue box represents the real object detected, the red box represents the missed object, and the green box represents the false alarm.

TABLE VI

COMPARISON OF DETECTION SPEED OF DIFFERENT METHODS. TEST ON THE MASATI DATASET WITH 512 × 512 IMAGE SIZE. THE BEST VALUE IS MARKED IN BOLD

Method	Params(Inference)	GFLOPs(Inference)	FPS	AP_{50}
EESRGAN-Det	64.00M	7705.2	0.7	74.07%
AERIS	171.23M	548.3	1.5	79.18%
DRENet	**4.79M**	8.4	**121.6**	77.60%
EfficientNetV2s	22.13M	8.8	51.7	82.70%
YOLOv8s6(Baseline)	9.90M	**8.2**	93.5	78.39%
Ours	9.90M	130.7	8.3	**83.03%**

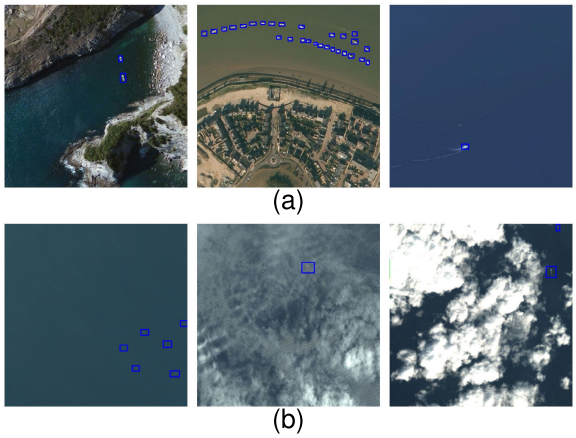


Fig. 7. Some examples of detection results on two public datasets obtained by the proposed method. (a) Results on MASATI. (b) Results on LEVIR-Ship.

TABLE VII

DETAILS OF INFERENCE TIME CONSUMED. THE AVERAGE RESULT OF ALL IMAGES IN MASATI IS DISPLAYED

	Generator	Detector	Decode & NMS	Total
Time(ms)	68.13	38.89	13.46	120.48
Proportion	56.55%	32.28%	11.17%	100%

EESRGAN-Det markedly trails behind other methods. This suggests that supervised image-level SR methods tend to yield satisfactory results solely on manually downsampled datasets containing paired HR-LR data. In contrast, our feature-level SR method exhibits remarkable generalization capabilities.

Fig. 7 displays some detection results of our method on the two datasets. The results indicate that our method still maintains a certain degree of robustness and can identify and locate ship targets accurately in scenes with complex land backgrounds and cloud occlusion.

2) Speed: The experiments were conducted on the MASATI dataset with an image size of 512 × 512. Table VI presents a comparison of detection speed among the aforementioned methods. In contrast to the YOLOv8s6 (baseline), our model maintains unchanged inference parameters since the sole disparity lies in a bilinear interpolation upsampling layer during inference. Consequently, our model exhibits significantly higher FLOPs (inference). The method implements a fourfold interpolation upscaling on the input image, escalating the number of input pixels to 16 times the original, thereby increasing FLOPs approximately 16 times, from 8.2 to 130.7 G. Nevertheless, in comparison with the image-level SR-based model (EESRGAN-Det), our model demonstrates considerably fewer FLOPs and a distinct advantage in FPS. These findings signify a balance achieved by our model between detection accuracy and speed: while

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TABLE VIII
ABLATION STUDY OF THE MODIFICATIONS IN THE BASELINE. THE OPTIMAL VALUES ARE MARKED IN BOLD

Baseline	Upsample Layer	Pixel Loss	Background Degrade	GAN Loss	AP_{50}
✓					90.84%
✓	✓				93.55%
✓	✓	✓			93.81%
✓	✓	✓	✓		94.90%
✓	✓	✓		✓	94.02%
✓	✓	✓	✓	✓	**95.17%**

TABLE IX
DETAILS OF ACCURACY ACROSS DIFFERENT TARGET SIZES. THE AVERAGE RESULT OF ALL IMAGES IN SHIP-LR IS DISPLAYED

Size Category	Target Number	AP_{50}	$AP_{50:95}$
Small	1516	92.09%	64.54%
Medium	490	95.23%	75.82%
Large	287	98.01%	90.81%

TABLE X
COMPARISON OF DIFFERENT INTERPOLATION. TEST ON MASATI. THE TIME REPRESENTS THE AVERAGE TIME CONSUMED BY UPSAMPLING ONE IMAGE. THE OPTIMAL VALUES ARE MARKED IN BOLD

Interpolation	AP_{50}	Time(ms)
Nearest-Neighbor	83.13%	**0.05**
Bilinear	83.29%	0.18
Bicubic	**83.30%**	0.31

TABLE XI
COMPARISON OF DIFFERENT UPSAMPLE SCALES. TEST ON MASATI. THE OPTIMAL VALUES ARE MARKED IN BOLD

Scale		AP_{50}	FPS
Training	Testing		
2	2	80.36%	**31.6**
	4	72.50%	8.3
4	2	70.13%	31.6
	4	**83.30%**	8.3

attaining heightened accuracy, the inference parameters remain unchanged, thereby maintaining a consistent detection speed.

Table VII illustrates the time allocation for each component during the inference of our model. The generator accounts for the majority of the total inference time, consuming 56.55%, as its primary tasks encompass feature extraction and dimensionality reduction. Meanwhile, the detector and postprocessing stages (decode and NMS) consume 32.28% and 11.17% of the total time, respectively.

D. Controlled Experiment

1) Ablation Experiments: We have implemented several enhancements to the original YOLOv8s6 prototype to improve small-ship detection performance. To assess the impact of these modifications, we conducted a series of ablation studies on our model, the results of which are presented in Table VIII. Several observations can be drawn from the results. First, the inclusion of an upsample layer at the backbone's input led to a significant 2.71% rise in detection AP. This improvement can be attributed to the interpolation upsample, which diminishes the scale discrepancy between small and large objects, facilitating the extraction of more discriminative features for smaller objects. Although these features may be somewhat blurred in comparison to the ground-truth HR feature, they still contribute significantly to feature expression. Second, the addition of pixel loss resulted in an increased detection AP. This highlights the positive impact of direct supervision signals in enhancing detection accuracy. Lastly, both the background degradation method and the GAN loss contributed to the enhancement of detection AP. We speculate that the former guided the generator to emphasize ship-related features over background elements, while the latter facilitated the creation of more realistic SR features, ultimately benefiting small-ship detection.

2) Performance Across Different Target Sizes: To verify the effectiveness of our proposed method, we also conducted experiments across different target sizes. Due to the greater diversity of target sizes within the Ship-LR dataset, we further categorized its small targets (sizes less than 1000 pixels) into three subgroups: small, medium, and large, with sizes ranging from less than 300 pixels, 300 pixels to 600 pixels, and above 600 pixels, respectively. Subsequently, we tested our method on the Ship-LR dataset and separately recorded the accuracy for the aforementioned three categories of targets. According to the results shown in Table IX, our method's performance remains relatively high even when the target sizes are tiny, demonstrating its effectiveness in small-ship detection tasks.

3) Different Upsample Settings: The layer performs bicubic interpolation on the input image, requiring no learnable parameters and offering flexibility to alter the interpolation method and upsample scales easily. Table X presents a comparison of various interpolation methods. The time metric denotes the average time taken to upsample an individual image. Interestingly, employing the relatively slower bicubic interpolation method appears to enhance detection accuracy. However, in general, different interpolation methods do not notably impact either the accuracy or speed of detection results. Furthermore, Table XI compares various upsample scales using bicubic interpolation. Observably, when the upsampling factor during training matches that during testing, higher detection accuracy is achieved with larger scales. However, this increase in accuracy is accompanied by a considerable decline in detection speed due to the quadratic relationship between calculation and scale. On the contrary, when the upsampling factor differs between training and testing, there is a significant reduction in detection accuracy. This discrepancy may be linked to the baseline model, which adapts to the true length instead of the relative length on the feature maps.

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V. Conclusion

In this article, we propose a novel HRFG-based method for small-ship detection on optical RS images. The proposed network performs four parts, including an HRFE to extract the HR feature as a direct supervisory signal, an HRFG to extract the SR feature from LR images directly, a binary classification discriminator to train the generator, and a detector to efficiently detect ships. The HRFE and discriminator are inference-free, thus our network is exactly the same as the baseline except for an extra upsampling layer at the input in the inference phase. Experiments on a self-assembled dataset and two public datasets verify that our proposed method has higher detection accuracy and ensures real-time detection speed at the same time. In our future work, we may consider introducing unsupervised SR methods to avoid upsampling of features and improve detection speed further.

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